

LAB 12: AI Q&A BOT WITH SEMANTIC SEARCH

Advanced NLP Pipeline using FAISS and MiniLM

1. Project Objective

Is project ka maqsad ek advanced **Question & Answering Bot** banana hai jo keyword matching ke bajaye **Semantic Meaning** (matlab) ko samajh kar jawab de sake. Humne is mein "Aesthetic Styling" ka dataset use kiya hai.

2. Technical Architecture

Component	Technology / Library
Language Model	Hugging Face (all-MiniLM-L6-v2)
Vector Database	FAISS (Facebook AI Similarity Search)
Backend Framework	Flask (Python)
Embeddings	Sentence-Transformers

3. NLP Pipeline Steps

Step 1: Data Preprocessing

Humne raw text data (Questions & Answers) ko structured Python lists mein convert kiya takay model use process kar sake.

Step 2: Vector Embeddings

Hugging Face ka **MiniLM** model use karte hue humne har sawal ko ek 384-dimensional numerical vector mein convert kiya. Isay "Text-to-Vector" transformation kehte hain.

Step 3: FAISS Indexing

In vectors ko fast search ke liye **FAISS Index** mein store kiya gaya. FAISS humein milliseconds mein sabse zyada miltay jultay (similar) vector nikaal kar deta hai.

Step 4: Semantic Retrieval

Jab user koi sawal puchta hai, bot us sawal ka vector banata hai aur FAISS index mein similarity search (L2 Distance) perform karta hai.

4. Core Code Implementation

```
# Similarity Search Logic
query_vector = model.encode([user_query])
distances, indices = index.search(np.array(query_vector), k=1)
if distances[0][0] < 1.5:
    reply = answers[indices[0][0]]
else:
    reply = "I'm not sure about that specific vibe."
```

5. Conclusion & Results

System successfully un sawalon ko bhi samajh raha hai jo dataset mein exact mojud nahi hain (e.g., matching 'informal' with 'minimalist'). Yeh project demonstrate karta hai ke kaise vector databases real-world AI applications mein kaam karte hain.